



**Analysing Machine Learning and Anomaly Detection Techniques for Credit Card
Fraud Detection**

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Declaration

‘I declare that this dissertation that I have submitted to Dublin Business School for the award of Master of Science in Business Analytics is the result of my own investigations, except where otherwise stated, where it is clearly acknowledged by references. Furthermore, this work has not been submitted for any other degree’.

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Abstract

Credit card fraud detection has become increasingly important due to the rapid growth of digital financial transactions and the rising sophistication of fraudulent activities. This paper explores the use of machine learning and anomaly detection algorithms to detect fraudulent transactions in a highly imbalanced credit card dataset. The research is conducted in the CRISP-DM method and uses data preprocessing methods such as feature scaling and Synthetic Minority Oversampling Technique (SMOTE). Various models were built and tested such as Logistic Regression, Random Forest, XGBoost, LightGBM, Multi-Layer Perceptron (MLP), Autoencoder, Hybrid Autoencoder-LightGBM model and Voting Ensemble model. The precision, recall, F1-score and ROC-AUC were used to evaluate the model performance. The outcomes exhibit that supervised ensemble approaches including random forest and voting ensemble had the best balanced performance in fraud detection, whereas the methods of anomaly detection had a higher rate of false positives. The study demonstrates the usefulness of ensemble learning methods in detecting frauds in highly imbalanced financial transactional data.

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List of Terms and Abbreviations

Abbreviation Full Form

ANN	Artificial Neural Network
AUC	Area Under the Curve
CNN	Convolutional Neural Network
CRISP-DM	Cross-Industry Standard Process for Data Mining
EDA	Exploratory Data Analysis
FPR	False Positive Rate
GAN	Generative Adversarial Network
KDD	Knowledge Discovery in Databases
LightGBM	Light Gradient Boosting Machine
LR	Logistic Regression
MLP	Multi-Layer Perceptron
PCA	Principal Component Analysis
PR-AUC	Precision–Recall Area Under the Curve
ReLU	Rectified Linear Unit
RF	Random Forest
RNN	Recurrent Neural Network
ROC	Receiver Operating Characteristic
ROC-AUC	Receiver Operating Characteristic – Area Under Curve
SMOTE	Synthetic Minority Oversampling Technique
TPR	True Positive Rate
XGBoost	Extreme Gradient Boosting

Chapter 1: Introduction

Digital payment systems have revolutionized the financial industry, making transactions quicker and more convenient. The speed at which this has developed has also introduced a rise in fraudulent activities, especially in the field of credit card transactions. Card fraud is a significant fraud risk for financial institutions and consumers, causing financial losses and undermining consumer confidence in electronic payments. The dynamic nature of fraud patterns and the highly imbalanced distribution of transaction data, fraudulent transactions are a small proportion compared to the total amount of transactions which makes it difficult to detect fraudulent transactions.

The traditional fraud detection systems that are based on a set of rules and manual checks are becoming less effective at detecting complex and evolving fraudulent activity. The systems are unable to identify the subtle irregularities and have high false positive rates. Machine learning techniques, on the other hand, offer a data-driven method that can identify hidden patterns in vast transaction datasets. Machine learning models can be used to enhance fraud detection systems and improve their accuracy and efficiency by using their statistical learning and pattern recognition capabilities. These models, however, are affected by a number of factors such as class imbalance, feature representation, and model selection.

This project will deal with application of machine learning techniques to detect credit card fraud with a real-world transaction dataset. The main goal of the research is to build and test predictive models that can correctly classify transactions as fraud or legitimate. To accomplish this, it is necessary to analyse the data set, preprocess the data, handle class imbalance and use various machine learning and anomaly detection models. These models are tested through appropriate metrics, such as precision, recall, F1-score and ROC-AUC, focusing on the correct identification of fraudulent transactions.

The research is quantitative and data-centric, uses a systematic and repeatable data mining methodology. The result of this project is an analytical artefact which will be a fraud detection model for support in decision making in financial transaction monitoring. Machine learning is a powerful tool that can be effectively used in real-world fraud detection situations and although full deployment is not the subject of this study, the proposed solution provides a good example of this.

1.1 Overview

Financial services is an industry plagued by fraud, which makes it a complex analytical challenge with a growing volume of transactions that are tough to detect in real time. Fraud is especially difficult to detect, as it is very rare compared to valid transactions, and the class imbalance can make the application of traditional predictive techniques less effective. This is why machine learning is playing an important role in fraud detection research, as it enables the analysis of massive amounts of transactional data to identify patterns and suspicious activities.

This project aims to determine whether the supervised machine learning and anomaly detection techniques can be used to make the job of fraud detection better in real European credit card transaction data. The study concentrates on features of the transactions, such as anonymised variables, transaction time, and transaction amount, and uses the preprocessing methods of scaling and SMOTE to overcome the class imbalance problem. The goal is to create a practical artefact which can help in fraud screening and identify the most appropriate methods for skewed financial data, plus to build a predictive model. This aligns with the Applied Research Project criterion which required the report to be research oriented and to create a 'practical output or artefact'.

The report is structured so that it takes the reader from the problem context to analysis and evaluation. The literature review is presented in Chapter 2, covering the subject areas of fraud detection and imbalance handling, supervised learning and anomaly detection. The methodology, data set, data pre-processing, model selection, and assessment method are described in Chapter 3. The results of the implementation and the comparison of the implemented models are discussed in Chapter 4. The findings, limitations and implications of this study are discussed in Chapter 5 and the directions for future studies are given at the end of the chapter.

1.2 Problem Statement

Fraudulent transactions are infrequent, hard to detect and predict, and a common problem in digital financial systems. Manual monitoring and rule-based fraud detection methods are becoming ineffective because of the huge number of transactions conducted online, and the fact that only a small percentage of them are fraud attempts.

While machine learning has been extensively applied to fraud detection, many of the current methods still struggle to reliably detect minority class fraud. Some models perform well on the overall accuracy but perform poorly on fraud identification, which reduces its usability in financial applications. This establishes a requirement for a more structured analytical approach that is able to evaluate multiple modelling techniques, deal with imbalance appropriately and generate an artefact that is useful for transaction monitoring and fraud screening. The research work is thus directed towards addressing the issue of maximizing the effectiveness of fraud detection on a highly imbalanced dataset using ML and anomaly detection techniques.

1.3 Research Question

Q1: What is the effectiveness of machine learning, anomaly detection and hybrid modelling approaches to detect credit card fraud in a highly imbalanced dataset, and which model can offer the best fraud detection accuracy whilst minimising the false positive rate?

1.4 Research Aim and Objectives

This research is aimed at creating and testing a machine learning method for credit card fraud detection using a real-world credit card transaction database. The research aims at finding the best modelling method to detect fraudulent transactions and how the various choices in the preprocessing affect the modelling performance.

The aim of the study is to analyse the structure of the data set, prepare the data for analysis, apply some of the most popular machine learning and anomaly detection models, Evaluate them based on relevant metrics for imbalanced classification. The final product of the project shall be a working artifact provided by creation of a fraud detection model that can be used for business decisions making in a financial context.

1.5 Research Rationale

The motivation for the research is that the detection of fraud in financial transactions demands techniques that are correct, flexible and applicable to highly skewed data. Machine learning approaches have been used successfully in this specific field with existing studies demonstrating the efficacy of various machine learning models, however this is dependent on a range of pre-processing, class balancing and feature representations. The literature review should therefore provide a demonstration of how the present study builds on previous studies and also, how this project addresses a gap in the literature.

Chapter 2: Literature Review

2.1 Introduction

Credit card fraud detection has become a critical area of research due to the rapid growth of digital financial transactions and the increasing sophistication of fraudulent activities.

Detecting fraudulent transactions is particularly challenging because of the highly imbalanced nature of transaction datasets, where fraudulent cases represent only a small fraction of the total data. As a result, traditional rule-based systems often fail to identify complex and evolving fraud patterns. Machine learning and deep learning approaches have therefore been widely explored to improve detection accuracy and efficiency. This chapter reviews existing literature on credit card fraud detection, focusing on machine learning models, deep learning techniques, data imbalance handling methods, and anomaly detection approaches.

2.2 Machine Learning Approaches for Fraud Detection

Several studies have explored the use of traditional machine learning algorithms for fraud detection. Alfaiz and Fati (2022) proposed an enhanced framework evaluating nine machine learning algorithms, including Logistic Regression, Random Forest, and CatBoost, using stratified cross-validation. Their findings showed that combining the AllKNN undersampling technique with CatBoost achieved superior performance, highlighting the importance of both model selection and resampling strategies.

Similarly, Khalid et al. (2024) demonstrated that ensemble-based machine learning models outperform individual classifiers in detecting fraudulent transactions, achieving improved balance between precision and recall. Jovanovic et al. (2022) further emphasised the importance of optimisation techniques by applying a Group Search Firefly Algorithm to tune machine learning models, significantly improving detection performance.

Earlier work by Varmedja et al. (2019) showed that classical models such as Random Forest and Logistic Regression can still achieve strong performance when combined with appropriate preprocessing techniques such as SMOTE. Taha and Malebary (2020) also demonstrated that optimised gradient boosting models, particularly LightGBM, can outperform several baseline classifiers when hyperparameters are carefully tuned.

While these studies confirm the effectiveness of traditional machine learning methods, many rely heavily on a single dataset and may not generalise well across different financial environments.

2.3 Hybrid and Ensemble Models

Hybrid and ensemble approaches have been proposed to improve fraud detection performance by combining the strengths of multiple models. Malik et al. (2022) introduced a two-phase hybrid architecture combining Adaboost with other classifiers, achieving improved performance compared to individual models. Esenogho et al. (2022) proposed a neural network ensemble using LSTM and AdaBoost, which effectively captured temporal dependencies in transaction data.

Mim et al. (2024) developed a soft voting ensemble approach that combined multiple classifiers and demonstrated statistically significant improvements in recall and precision. These findings suggest that ensemble learning can enhance detection performance, particularly in imbalanced datasets.

However, ensemble models often increase computational complexity and may be difficult to deploy in real-time systems without optimisation.

2.4 Deep Learning and Advanced Architectures

Deep learning techniques have gained significant attention in fraud detection due to their ability to model complex patterns. Alharbi et al. (2022) proposed a CNN-based approach using a text2IMG mechanism to transform transaction data into images, achieving high accuracy but relatively lower F1-scores due to class imbalance.

Alarfaj et al. (2022) demonstrated that deeper CNN architectures can outperform traditional machine learning models, achieving high accuracy and improved fraud detection performance. Yu et al. (2024) further advanced this area by integrating GANs with RNN architectures, achieving near-perfect detection results.

Recent studies by Mienye and Sun (2023a) and Mienye and Swart (2024) explored hybrid deep learning approaches combining autoencoders with gradient boosting models, demonstrating strong performance on highly imbalanced datasets. Transformer-based models have also been introduced, achieving superior performance and generalisation across datasets.

Despite their strong performance, deep learning models typically require high computational resources and large amounts of data, which may limit their practical deployment.

2.5 Anomaly Detection Approaches

Anomaly detection techniques provide an alternative approach by identifying deviations from normal transaction behaviour. Jiang et al. (2023) proposed an unsupervised attentional anomaly detection network combining autoencoders and GANs, achieving strong performance without requiring labelled fraud data.

Lin and Jiang (2021) introduced a hybrid approach combining autoencoders with probabilistic random forests, demonstrating stability even without resampling techniques. These methods are particularly useful in scenarios where labelled fraud data is limited.

However, anomaly detection methods may struggle to achieve high precision, as not all anomalies correspond to fraudulent transactions.

2.6 Imbalanced Data Handling and Data Augmentation

Handling class imbalance remains a critical challenge in fraud detection. Strelcenia and Prakoonwit (2023) proposed a GAN-based augmentation method (K-CGAN) that generates realistic synthetic fraud samples, significantly improving classification performance.

Dang et al. (2021) highlighted the importance of correct resampling procedures, showing that improper application can lead to data leakage and inflated results. Tingfei et al. (2020) demonstrated that variational autoencoders can effectively generate synthetic fraud samples, outperforming traditional methods such as SMOTE.

Although these techniques improve model performance, they may introduce synthetic bias if the generated data does not accurately represent real fraud patterns.

2.7 Feature Selection and Optimisation Techniques

Feature selection plays a key role in improving model efficiency and performance. Mienye and Sun (2023b) proposed a hybrid feature selection method combining Information Gain and Genetic Algorithms, achieving high sensitivity and AUC scores.

Du et al. (2023) explored the impact of combining feature extraction techniques such as PCA and autoencoders with sampling methods, showing that the order of preprocessing steps

significantly affects performance. Ayoub et al. (2023) demonstrated that optimisation techniques such as Particle Swarm Optimisation can enhance classification accuracy.

2.8 Emerging Techniques and Real-World Applications

Recent research has explored advanced techniques such as federated learning, transfer learning, and graph neural networks. Alarfaj and Shahzadi (2024) proposed a federated learning framework to address privacy concerns in distributed financial systems.

Mustafa Abdul Salam et al. (2024) applied graph neural networks for real-time fraud detection, demonstrating the ability to capture complex relationships between transactions. Lebichot et al. (2021) explored transfer learning approaches, enabling fraud detection models to adapt to new domains with limited labelled data.

While these approaches show strong potential, they often require complex implementation and significant computational resources.

2.9 Research Gap

Despite the wide range of approaches explored in the literature, several research gaps remain. Many studies focus on a limited set of models or rely on a single dataset, reducing the generalisability of their findings. Additionally, there is limited research that systematically compares traditional machine learning, deep learning, and anomaly detection approaches within a unified framework. Furthermore, while class imbalance handling techniques are widely used, their interaction with different model types is not always thoroughly evaluated.

Therefore, this study aims to address these gaps by evaluating multiple machine learning and anomaly detection techniques within a single framework, with a focus on handling class imbalance and improving fraud detection performance.

2.10 Summary

This chapter reviewed existing research on credit card fraud detection, highlighting the use of machine learning, deep learning, ensemble methods, and anomaly detection techniques. While significant progress has been made, challenges related to class imbalance, model generalisation, and computational complexity remain. The next chapter presents the methodology used in this study, outlining the research design, dataset, and modelling approach adopted to address these challenges.

Chapter 3: Research Methodology and Project Management

3.1 Introduction

This chapter outlines the methodology adopted in this study to develop and evaluate machine learning models for credit card fraud detection. It describes the data mining framework used, the dataset and its characteristics, the data preprocessing steps, and the modelling techniques implemented. Additionally, the chapter presents the project management approach, including task planning, timeline, and risk management strategies.

3.2 Methodology Framework

Data analytics and machine learning projects are often organized to the well-known process frameworks like CRISP-DM (Cross Industry Standard Process for Data Mining) and the Knowledge Discovery in Databases (KDD) framework. These paradigms offer a systematic way of deriving meaningful information of data.

KDD process is made up of various stages that follow one another such as data selection, preprocessing, transformation, data mining and interpretation. It is more concerned with the technical operation of extracting knowledge on datasets. CRISP-DM on the other hand is a more detailed and iterative process that consists of six steps, namely business understanding, data understanding, data preparation, modelling, evaluation, and deployment. CRISP-DM pays more attention to the correspondence of data analysis to business goals and practical uses.

3.3 Methodology Used

The CRISP-DM framework is taken as the methodological ground. CRISP-DM has been chosen because it is systematic and iterative and that fits the purpose of detection of fraud based on machine learning methods.

Contrary to the KDD process, CRISP-DM explicitly embraces business knowledge and deployment factors and is more applicable in practical financial analytics issues. Also, its iterative nature enables the models and preprocessing methods to be refined on-the-fly and this is necessary in cases where the data is highly imbalanced like credit card transaction data. Thus, CRISP-DM is a framework that offers a viable and industry-current method of creating and appraising fraud detection models.

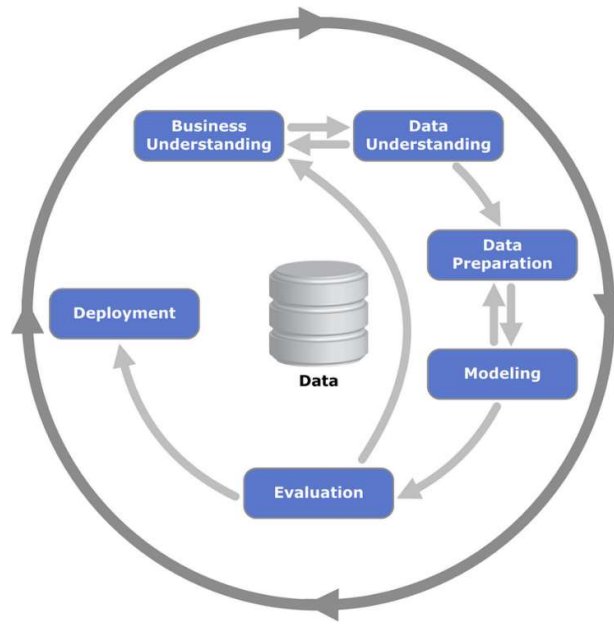


Figure 3.1: CRISP-DM Framework

3.4 Application of CRISP-DM

3.4.1 Business Understanding

The objective of this project is to develop a machine learning-based system for detecting fraudulent credit card transactions. Fraud detection is a critical problem in financial systems due to the increasing volume of digital transactions and the significant financial losses associated with fraudulent activities. The primary goal is to accurately identify fraudulent transactions while minimising false positives, ensuring the practical applicability of the model.

3.4.2 Data Understanding

The dataset used in this study is the European cardholders' credit card transaction dataset, which contains anonymised transactions made by cardholders over a two-day period in September 2013. The dataset is widely used in fraud detection research and consists of 284,807 transactions, of which only 492 are fraudulent, representing approximately 0.172% of the total data. Due to confidentiality constraints, most features have been transformed using Principal Component Analysis (PCA), resulting in 28 anonymised variables labelled V1 to V28. In addition to these, the dataset includes 'Time', representing the seconds elapsed between each transaction and the first transaction, and 'Amount', which indicates the monetary value of each transaction. The target variable 'Class' indicates whether a

transaction is fraudulent (1) or legitimate (0). The extreme class imbalance and anonymised feature space make this dataset particularly suitable for evaluating machine learning and anomaly detection techniques.

3.4.2.1 Dataset Structure Table

Attribute Name	Data Type	Range of Values	Unit	Description
Time	Numeric (Float)	Positive values	Seconds	Time elapsed between each transaction and the first transaction in the dataset.
V1 – V28	Numeric (Float)	Continuous (Positive and negative values)	N/A	Anonymised features obtained using PCA transformation to protect sensitive information.
Amount	Numeric (Float)	Positive values	Euros	Transaction amount.
Class	Categorical (Binary)	{0,1}	N/A	Target variable (0 = legitimate transaction, 1 = fraudulent transaction).

Table 3.1: Dataset Structure

3.4.2.2 Dataset Visualisation

I. Class Distribution

The following bar chart shows the distribution of the target variable (Class), where 0 represents legitimate transactions and 1 represents fraudulent transactions. Fig.1 clearly indicates a severe class imbalance, with fraudulent transactions forming only a very small fraction of the dataset compared to legitimate ones. This imbalance is a critical challenge in fraud detection, as machine learning models may become biased toward predicting the majority class. Therefore, techniques such as resampling, class weighting, or anomaly detection approaches are necessary to ensure the model effectively identifies fraudulent cases. This extreme imbalance indicates that standard accuracy is not a reliable evaluation metric, making metrics such as precision, recall, F1-score, and AUC more appropriate for assessing model performance.

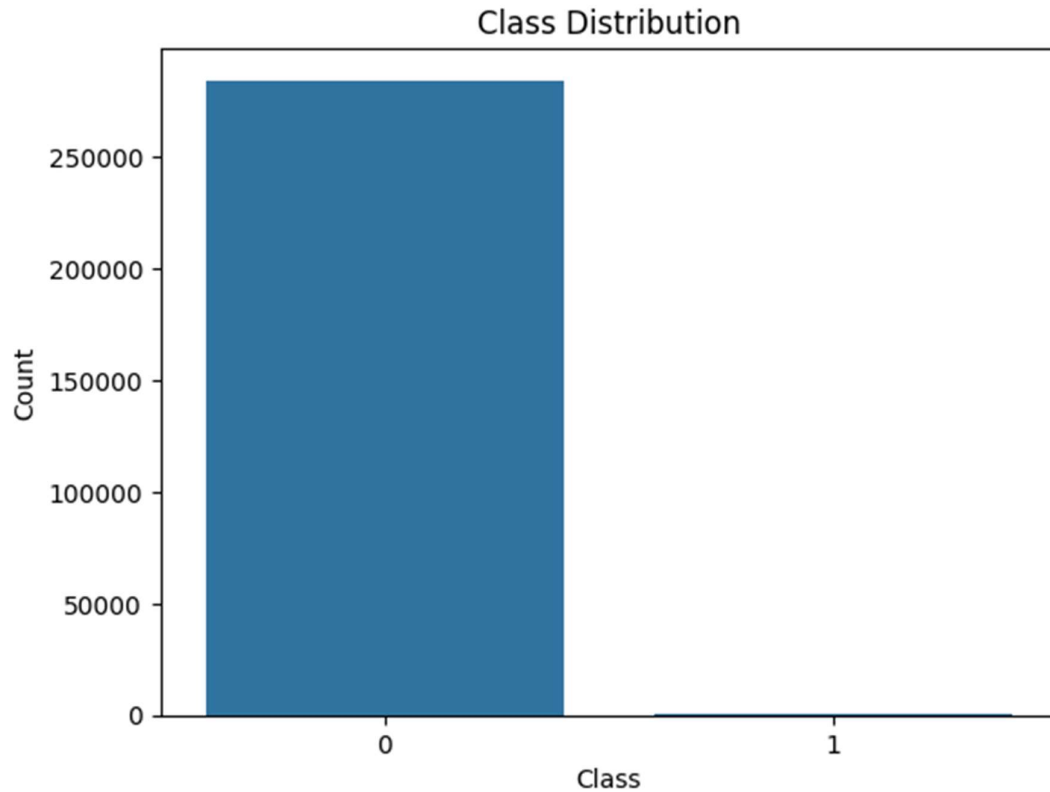


Figure 3.2: Class Distribution

II. Transaction amount Distribution

The following histogram (Fig.2) illustrates the distribution of transaction amounts, where the y-axis represents the frequency (number of transactions) within each value range. The data is highly right-skewed, with most transactions occurring at low values and a small number of transactions having very large amounts. This indicates the presence of outliers and heavy-tailed behaviour, which is common in financial data.

Such skewness can negatively affect certain machine learning models, so transformations like log-scaling or normalization may be required during preprocessing to improve model performance and stability.

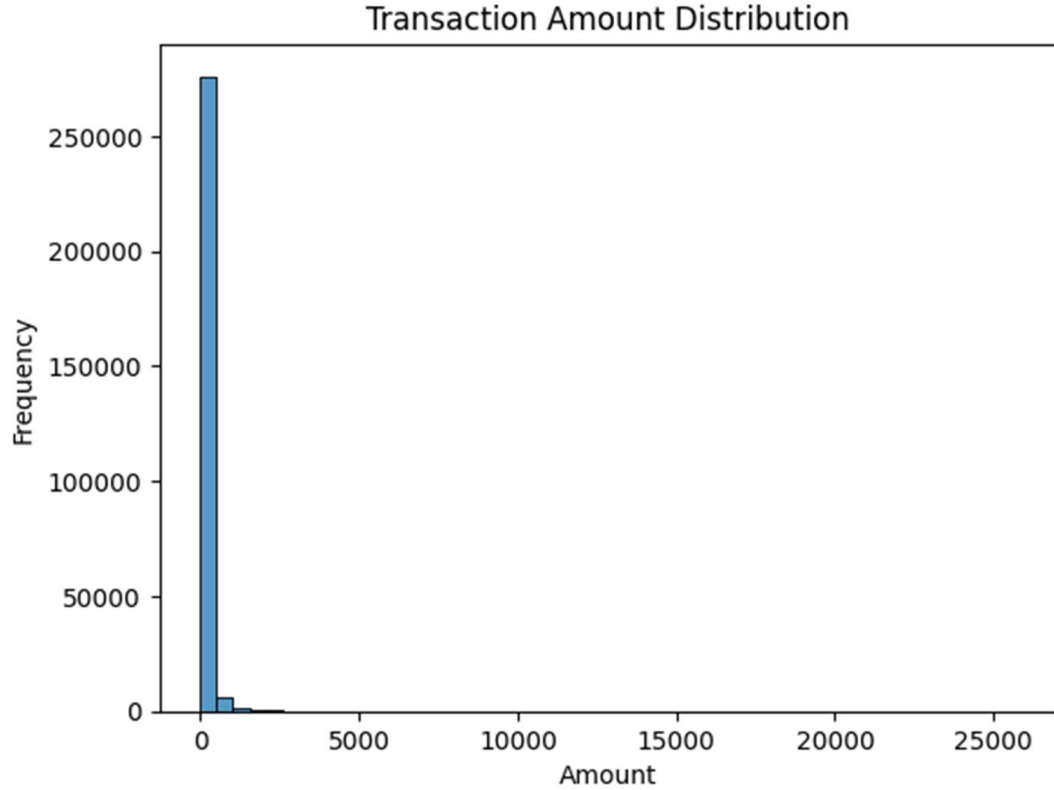


Figure 3.3: Transaction Amount Distribution

III. Transaction Time Distribution

The following histogram (Fig.3) shows how transactions are distributed over time, where the ‘time’ feature represents the elapsed seconds since the first transaction in the dataset. The distribution reveals that transactions are not uniformly distributed, with certain periods showing higher activity. This suggests underlying temporal patterns in user behaviour. However, since the time variable is relative, it does not directly correspond to real-world timestamps (e.g., hours or days). Despite this, time-based patterns can still be useful for modelling, particularly in detecting unusual transaction sequences or bursts of activity, making it relevant for anomaly detection and time-aware models.

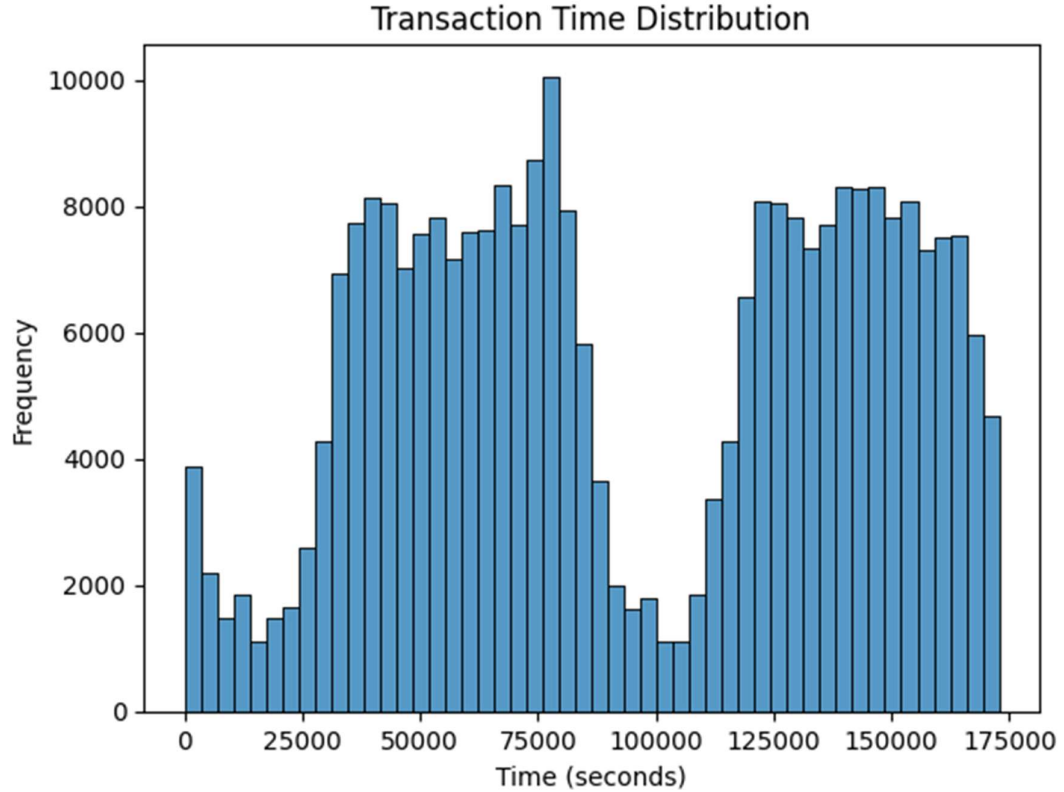


Figure 3.4: Transaction Time Distribution

IV. Feature Correlation Heatmap

The correlation heatmap (Fig.4) displays the relations among all variables in the dataset, including the anonymized variables (V1-V28), Time, Amount and Class. The majority of the features have a low correlation between them, which is anticipated because PCA is used to derive the V1-V28 variables and thus, they are orthogonal. Moreover, none of the features has a strong linear relationship with the target variable (Class), indicating that fraud detection in this data is a non-linear and complex problem. This is a reason to use advanced machine learning models (e.g., ensemble methods or neural networks e.g. Random Forest, XGBoost, Neural Networks), rather than using simple linear classifiers.

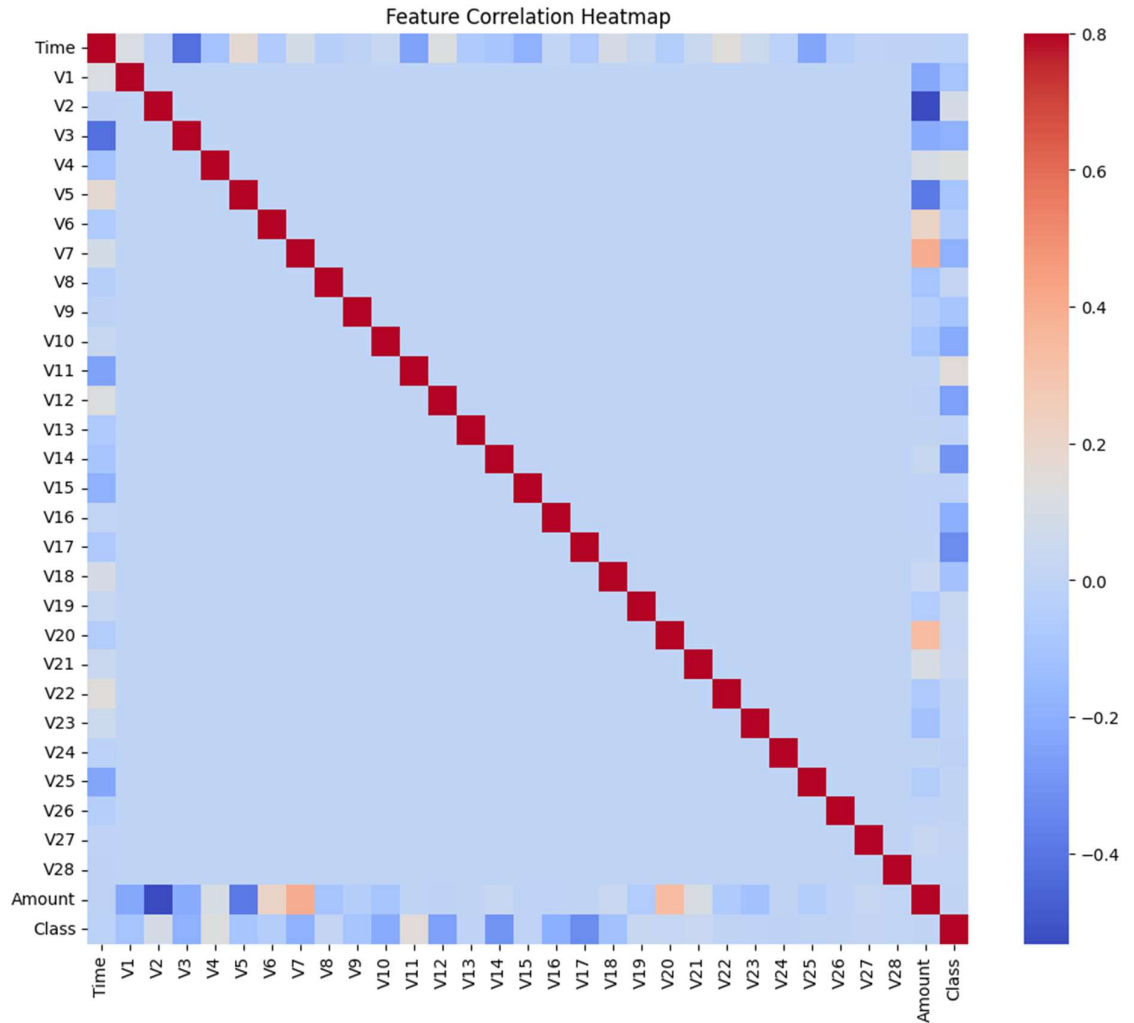


Figure 3.5: Feature Correlation Heatmap

V. Amount vs Class Boxplot

The boxplot (Fig.5) compares between the distribution of the amount of transactions in legitimate (Class = 0) and fraudulent (Class = 1) transactions. It demonstrates that the fraudulent transactions are more likely to be distributed differently with a lower median and a greater spread and variability. This means that fraudsters can strive to evade detection through small or less noticeable transactions, but sometimes conduct larger anomalous transactions. This phenomenon of overlapping distributions also implies that transaction amount alone cannot be used to classify transactions, although this feature is still important in a machine learning model when combined with other features.

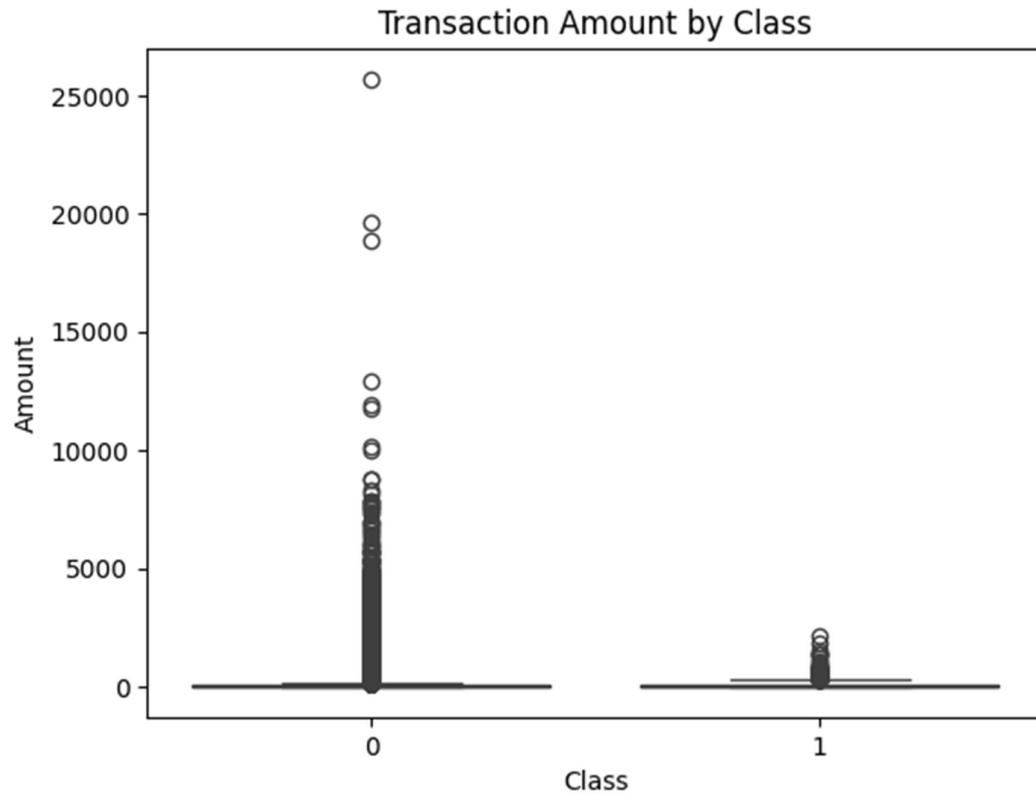


Figure 3.6: Amount vs Class Boxplot

Exploratory Data Analysis (EDA) was carried out to learn more about the dataset and find out the most important patterns to use in fraud detection. It was found that there was a large imbalance in the classes, as the proportion of fraudulent transactions was very low in the dataset. Such an imbalance points to the necessity of specialised methods like resampling and anomaly detection.

The distribution of the amount of transactions was identified to be extremely skewed, with most values of transactions being closer to lower values, and a few outliers of large value. This implies that the scaling methods are required to normalise the feature in order to make the model effective. The correlation analysis revealed that the majority of the PCA-transformed features have low linear relationships with others which is understandable in the context of the PCA transformation. Nevertheless, some features demonstrated rather high associations with the target variable, which means that they might be crucial in fraud detection. In general, the exploratory analysis was very informative and guided the process of data preprocessing and the choice of the right machine learning and anomaly detection methods applied in this work.

3.4.3 Data Preparation

Data preparation is a critical phase in the CRISP-DM framework, as the quality of preprocessing directly influences model performance and reliability. Based on the findings from the exploratory data analysis, several preprocessing steps were carried out to ensure that the dataset was suitable for machine learning and anomaly detection modelling. These steps included data cleaning, feature scaling, dataset partitioning, and class imbalance handling.

3.4.3.1 Data Cleaning and Validation

The dataset was initially examined for missing values, duplicate records, and inconsistencies that could affect model training. A comprehensive validation check was conducted to ensure data completeness and integrity.

The analysis confirmed that the dataset contained no missing values across any of the features, eliminating the need for imputation techniques. Duplicate records were also assessed to ensure that repeated transactions did not introduce bias into the training process. No significant duplicate-related issues were identified.

Data validation was further performed by checking feature distributions and confirming that all variables were correctly represented in their expected numeric format. This step ensured that the dataset could be processed directly without requiring additional structural transformation.

3.4.3.2 Feature Scaling

Feature scaling was applied to normalise the numerical attributes and ensure consistency across feature ranges. Although the anonymised PCA-transformed variables (V1–V28) were already standardised as part of the original dataset preparation, the Time and Amount variables required additional scaling.

Feature scaling was particularly necessary because several models used in this study, including Multi-Layer Perceptron and Autoencoder, are highly sensitive to differences in feature magnitude. Without scaling, features with larger numeric ranges could dominate the optimisation process and negatively affect convergence.

The need for scaling was further supported by the exploratory analysis, which revealed that transaction amounts were highly skewed and exhibited significant variation.

3.4.3.3 Dataset Splitting

To evaluate model performance objectively, the dataset was divided into training and testing subsets using an 80:20 split.

The partitioning was performed using stratified sampling, ensuring that the class distribution of fraudulent and legitimate transactions was preserved in both subsets.

Stratification is particularly important in highly imbalanced datasets because random splitting may produce subsets with insufficient fraud samples, leading to unreliable evaluation results.

3.4.3.3 Handling Class Imbalance

One of the most significant challenges identified during data understanding was the extreme imbalance between legitimate and fraudulent transactions. Fraudulent transactions accounted for only 0.172% of the dataset, which creates a substantial risk of model bias toward the majority class. If left unaddressed, this imbalance would lead to misleadingly high accuracy while failing to detect fraud effectively. To address this issue, the Synthetic Minority Oversampling Technique (SMOTE) was applied to the training data. SMOTE was applied only to the training dataset to avoid data leakage. Applying oversampling before splitting would allow synthetic information to influence the test set, resulting in artificially inflated performance.

The use of SMOTE was directly justified by the severe imbalance observed during exploratory analysis and was essential for improving the ability of supervised learning models to identify fraudulent patterns.

3.4.4 Modelling

A range of machine learning and deep learning models are implemented to classify transactions as fraudulent or legitimate. These include Logistic Regression as a baseline model, Random Forest as a classical ensemble method, and gradient boosting models such as XGBoost and LightGBM for high-performance classification. In addition, a Multi-Layer Perceptron (MLP) is implemented to incorporate a neural network approach. An Autoencoder is also used as an unsupervised anomaly detection model, trained to identify deviations from

normal transaction behaviour. Also, a soft voting ensemble model using predictions from Random Forest and XGBoost is used.

3.4.4.1 Logistic Regression

Logistic Regression is a fundamental supervised learning algorithm used for binary classification problems. It serves as a baseline model in this study due to its simplicity and interpretability. The model estimates the probability that a given transaction belongs to the fraudulent class based on a linear combination of input features.

The relationship between the input variables and the output is modelled using the logistic (sigmoid) function:

$$P(y = 1 | x) = \frac{1}{1 + e^{-z}}$$

where:

$$z = \beta_0 + \beta_1 x_1 + \dots + \beta_n x_n$$

Here, β represents model coefficients and x represent input features.

The model outputs probabilities, which are converted into class labels using a threshold (typically 0.5). Logistic Regression assumes a linear decision boundary, which limits its ability to capture complex relationships in data.

Despite its simplicity, Logistic Regression is included in this study to provide a benchmark against which more advanced models can be compared. It is particularly useful for understanding how much performance improvement is achieved by more complex approaches.

3.4.4.2 Random Forest

Random Forest is an ensemble learning method that combines multiple decision trees to improve predictive performance and reduce overfitting. Instead of relying on a single tree, the model aggregates predictions from many trees trained on different subsets of the data.

The model uses bootstrap sampling, where each tree is trained on a randomly sampled subset of the dataset, and random feature selection, where only a subset of features is considered at each split. This introduces diversity among the trees and improves generalisation. Random Forest is particularly effective for fraud detection because it can capture non-linear

relationships, handle high-dimensional data, and is robust to noise and outliers. Its ensemble nature makes it well-suited for imbalanced datasets, especially when combined with resampling techniques such as SMOTE.

3.4.4.3 XGBoost

XGBoost is an advanced implementation of gradient boosting that builds models sequentially, where each new model corrects the errors of previous ones. It is widely used due to its high performance and efficiency.

XGBoost is particularly suitable for fraud detection because it focuses on misclassified samples during training, which improves its ability to detect minority-class fraud cases. Its strong recall performance makes it valuable in scenarios where missing fraudulent transactions is costly.

3.4.4.4 LightGBM

LightGBM is a gradient boosting framework designed for efficiency and scalability. It differs from traditional boosting methods by using histogram-based learning and a leaf-wise tree growth strategy.

The model follows the same boosting principle:

$$F(x) = \sum_{m=1}^M \gamma_m h_m(x)$$

but optimises performance through faster computation and reduced memory usage.

LightGBM grows trees by expanding the leaf with the highest loss reduction, which often leads to deeper and more complex trees compared to level-wise methods.

This model is suitable for this study due to its ability to handle large datasets efficiently and capture complex feature interactions. However, it may require careful tuning to avoid overfitting.

3.4.4.5 Multi-Layer Perceptron (MLP)

The Multi-Layer Perceptron is a type of artificial neural network that consists of multiple layers of interconnected neurons. It is capable of modelling complex, non-linear relationships in data.

Each neuron computes a weighted sum of inputs:

$$z = Wx + b$$

followed by an activation function:

$$a = f(z)$$

In this study, the ReLU activation function is used:

$$f(x) = \max(0, x)$$

The network is trained using backpropagation to minimise prediction error.

MLP is suitable for fraud detection because it can learn complex patterns that may not be captured by linear or tree-based models. However, it requires proper scaling of input data and careful tuning to avoid overfitting.

3.4.4.6 Autoencoder

An Autoencoder is an unsupervised neural network used for anomaly detection. It learns to reconstruct input data by compressing it into a lower-dimensional representation and then reconstructing it.

The model consists of:

- Encoder: $z = f(x)$
- Decoder: $\hat{x} = g(z)$

The reconstruction error is calculated as:

$$\text{Error} = ||x - \hat{x}||^2$$

Autoencoders are particularly useful for fraud detection because they are trained on normal transactions and can identify deviations from normal behaviour. However, they may struggle to achieve high precision since anomalies are not always fraudulent.

3.4.4.7 Hybrid Model (Autoencoder + LightGBM)

The hybrid model combines anomaly detection with supervised learning to improve fraud detection performance.

In this approach, the reconstruction error from the Autoencoder is added as a new feature:

$$X' = X \cup \{\text{AE error}\}$$

This enriched feature set is then used to train a LightGBM model.

The hybrid approach leverages the strengths of both methods:

- Autoencoder identifies unusual patterns
- LightGBM classifies transactions using labelled data

This makes the model more effective in capturing both known and unknown fraud patterns.

3.4.4.8 Voting Ensemble

The Voting Ensemble combines predictions from multiple models to improve overall performance and stability.

In soft voting, the final prediction is based on the average predicted probabilities:

$$\hat{y} = \arg \max \sum_{i=1}^n w_i P_i(y | x)$$

where:

- $P_i(y | x)$ is the predicted probability from each model
- w_i is the weight assigned to each model

In this study, Random Forest and XGBoost are combined due to their strong individual performance.

The ensemble approach reduces variance and improves robustness, making it suitable for fraud detection tasks where reliability is critical.

3.4.5 Evaluation

Model performance is evaluated using the test dataset. Due to the imbalanced nature of the data, multiple evaluation metrics are used, including precision, recall, F1-score, and the Area Under the Receiver Operating Characteristic Curve (ROC-AUC).

Special emphasis is placed on recall for the fraud class, as failing to detect fraudulent transactions can lead to significant financial loss. The evaluation process compares the performance of all models to determine their effectiveness and suitability for fraud detection.

3.4.6 Deployment

Although full deployment is beyond the scope of this project, the developed models demonstrate how machine learning techniques can be applied in real-world fraud detection

systems. The findings provide insights into how such models could be integrated into financial monitoring systems for near real-time fraud detection.

3.5 Project Management

Effective project management is essential to ensure that the research is completed within the allocated timeframe while maintaining quality and consistency. This project follows a structured approach to planning, execution, monitoring, and risk control. Given the iterative nature of machine learning projects, flexibility was incorporated into the planning process to allow refinement of models and preprocessing techniques based on intermediate results.

The project management approach aligns with the CRISP-DM framework, ensuring that each stage of the data mining process is systematically executed while allowing feedback loops between stages.

3.6 Project Tasks

The project is structured into several stages. Initially, research planning is conducted to define the problem, objectives, and methodology. This is followed by a literature review to identify existing approaches and research gaps. The dataset is then analysed through exploratory data analysis, followed by data preprocessing and model development. Finally, the models are evaluated, and the findings are documented in the final report.

3.7 Project Timeline

The project is carried out over a 12-week period. The initial weeks focus on research planning and literature review, followed by data understanding and preprocessing. Model implementation and evaluation are conducted in the middle phase, while the final weeks are dedicated to report writing, revision, and submission.

Task	Duration
Research Planning	Week 1
Literature Review	Weeks 1–3
Data Collection	Week 2
Data Preprocessing & Analysis	Weeks 3–5
Model Development	Weeks 5–8
Model Evaluation	Weeks 8–9
Report Writing	Weeks 9–12
Final Review & Submission	Week 12

Table 3.2: Project Timeline

3.8 Risk Management

Several potential risks are identified and addressed in this project. Hardware or system failure is mitigated through regular backups and the use of cloud-based platforms. The issue of class imbalance is addressed using SMOTE. Model underperformance is mitigated by implementing multiple models and comparing their results. Time constraints are managed through a structured project timeline, while data quality issues are handled through thorough preprocessing and validation.

Risk	Likelihood	Impact	Mitigation Plan
Hardware or system failure	Low	High	Perform regular backups and use cloud storage
Class imbalance affecting model performance	High	Medium	Apply resampling techniques such as SMOTE
Model underperformance	Medium	High	Experiment with multiple algorithms and tuning methods
Data limitations or quality issues	Medium	Medium	Perform thorough data preprocessing and validation

Table 3.3: Risk Management Plan

3.9 Ethical Considerations

The dataset selected for this study is a publicly available anonymised credit card sale data set, consisting of no personally identifiable details. The anonymisation of transaction attributes using PCA transformation guarantees the privacy and confidentiality of the cardholders. The data set is not sensitive but the privacy of the participants is not significantly at risk.

Yet, ethical issues are still salient in fraud detection studies. Financial systems utilizing machine learning models can generate false positives, impacting genuine customers with needless transaction restrictions or monetary inconvenience. For this reason, not only was accuracy evaluated but also precision and recall to provide even performance of fraud detection.

Financial decisions, with regard to the economic feasibility of the developed models, were not taken in the real world, and the study was done purely for academic and research purposes.

3.9 Summary

This section presented the project management approach adopted in the study. A structured plan was implemented to guide the research process, supported by a defined timeline, continuous monitoring, and risk management strategies. The integration of project management principles with the CRISP-DM framework ensured that the study was conducted systematically and efficiently.

Chapter 4: Implementation and Results

4.1 Introduction

This chapter presents the implementation and evaluation of the machine learning and anomaly detection models discussed in the previous chapter. The goal of the implementation phase is to use the models selected to build on the data set prepared to evaluate the models' performance in detecting fraudulent transactions in credit card usage. All the models were mapped with same pre-processing techniques to make the models comparable and model specific configurations and parameters were set as per the characteristics of the algorithm. The results are measures by performance metrics appropriate for highly imbalanced classification problem, such as precision, recall, F1-score and ROC-AUC.

4.2: Implementation Environment

The models have been run in Python on the Google Colab platform. During the implementation process several machine learning and deep learning libraries were utilized.

Library	Purpose
Scikit-learn	Machine learning models and evaluation
TensorFlow/Keras	Deep learning implementation
Imbalanced-learn	SMOTE resampling
Pandas	Data manipulation
NumPy	Numerical computation
Matplotlib & Seaborn	Data visualisation

The experiments were conducted using Google Colab due to its support for GPU acceleration and cloud-based computational resources.

4.3: Logistic Regression

Logistic Regression was used as a baseline supervised learning model as a reference for comparison with more complex models. It assumes a linear separating boundary in the model and predicts the probability of a transaction belonging to the fraud class. This model was

chosen due to its simplicity, interpretation, and widespread application in binary classification problems.

The data was preprocessed in the training data before the model was trained with feature scaling and SMOTE resampling. To ensure convergence during optimisation, the maximum number of iterations was increased.

Parameter	Value
max_iter	1000

Table 4.1: Logistic Regression Parameters

The precision score of the model was 0.06, the recall 0.92 and the F1 score 0.11 for the fraud class. Precision is very low thereby a lot of legitimate transactions are misclassified as fraud, whereas recall is high, and a lot of the fraudulent transactions are correctly classified. This leads to a high false positive rate, disqualifying the model from use in a real world environment even though it is able to identify fraud cases.

4.4: Random Forest

Random Forest was introduced as an ensemble model that could deal with complex non-linear relationships and interactions among the features. It uses the algorithm of creating several decision trees and utilizing majority voting for prediction, increasing the stability of classification and decreasing overfitting. This model was chosen because of its high accuracy on structured tabular data and its ability to cope with imbalanced classification problems. To ensure the reproducibility of the model, 100 decision trees were used for training, and random state was set to a fixed value.

Parameter	Value
n_estimators	100
random_state	42

Table 4.2: Random Forest Parameters

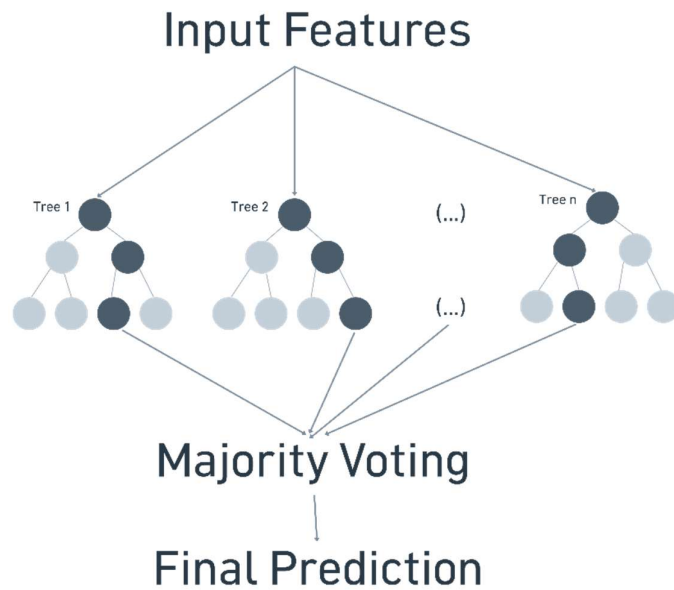


Figure 4.1: Random Forest Architecture

For the fraud class, the model had a precision of 0.85, a recall of 0.84 and an F1-score of 0.84. These findings suggest that the model has a good precision-recall balance, ensuring a reliable detection of fraudulent transactions. Random Forest is one of the models that performs well in all models tested.

4.5: XGBoost

To boost predictive accuracy, XGBoost was built as a gradient boosting ensemble algorithm to correct classification errors made by the previous trees. XGBoost is a popular model that is known to perform well on classification problems with structured data and imbalanced datasets.

Multiple estimators, tree depth control, and subsampling parameters were used in the model to increase generalisation, while minimising overfitting.

Parameter	Value
n_estimators	200
random_state	42
max_depth	6
learning_rate	0.1
subsample	0.8
colsample_bytree	0.8

Table 4.3: XGBoost Parameters

The model achieved a precision of 0.64, recall of 0.89 and an F1 of 0.74 for the fraud class. The high recall rate means the model is good at catching frauds, but the lower precision rate suggests that there is some overlap of genuine and fraudulent transactions. This means that in an environment where fraud detection is more important than reducing false alarms, XGBoost could be a good choice.

4.6: LightGBM

LightGBM is a high-performance gradient boosting framework that is optimized for training and scalability. LightGBM has adopted a leaf-wise tree growth strategy that makes it more efficient to train and faster when compared to traditional boosting algorithms.

A lower learning rate and multiple estimators were used to train the model in order to increase the stability of the classification and the generalisation.

Parameter	Value
n_estimators	200
random_state	42
learning_rate	0.05

Table 4.4: LightGBM Parameters

For the fraud class, the model had a precision of 0.49, a recall of 0.87, and an F1 score of 0.62. The recall is relatively high, but the lower precision means that the model is more likely

to identify transactions as fraudulent than actually commit fraud. This is less effective than other ensemble techniques.

4.7: Multi-Layer Perceptron (MLP)

Multi-Layer Perceptron (MLP) was designed as a feedforward Artificial Neural Network that could learn non-linear relationships that are too complex for linear learning methods. Neural networks are especially useful if the behavior of the transactions included in the model has complex patterns which are not easily captured by traditional linear models.

A two-layered network with the ReLU activation function was used for the network architecture.

Parameter	Value
hidden_layer_sizes	(64, 32)
activation	ReLU
max_iter	100

Table 4.5: MLP Parameters

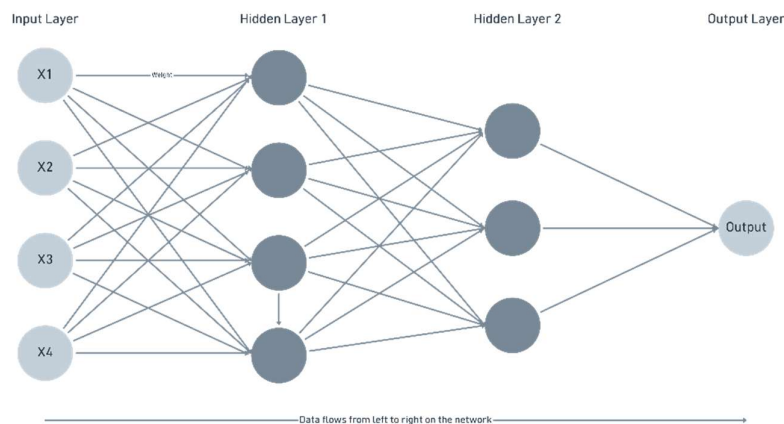


Figure 4.2: MLP Architecture

For the fraud class, the MLP had a precision of 0.76, recall of 0.84, and an F1-score of 0.80. The results show a high level of precision and recall, which suggests that the model is effective in detecting fraud and minimizing false positives. It performs as well as ensemble methods.

4.8: Autoencoder

The Autoencoder model was implemented as an unsupervised anomaly detection model, which reconstructs the normal transactions. High reconstruction errors were flagged as indicating an abnormal transaction and/or fraud. Fraud detection is one application that makes autoencoders very valuable since fraud is rare and could be very different from typical transaction patterns.

The model is trained with Mean Squared Error (MSE) loss and Adam optimiser.

Parameter	Value
epochs	50
batch_size	256
loss	MSE
optimizer	adam

Table 4.6: Autoencoder Parameters

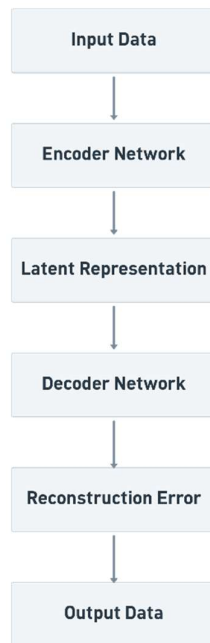


Figure 4.3: Autoencoder Architecture

The model achieved the results of 0.27 for precision, 0.83 for recall and 0.41 for F1-score. The model is good at detecting the majority of cases, however, due to the low precision there are a lot of false positive cases. This is a typical limitation of anomaly detection methods.

4.9: Hybrid Model (Autoencoder + LightGBM)

A hybrid fraud detection model was created by combining the reconstruction error obtained from the Autoencoder with the LightGBM classifier as an extra feature. This method is based on the idea of unsupervised anomaly detection and supervised ensemble learning to enhance fraud classification capability.

Component	Description
Base Model	LightGBM
Additional Feature	Autoencoder reconstruction error

Table 4.7: Hybrid Model Components

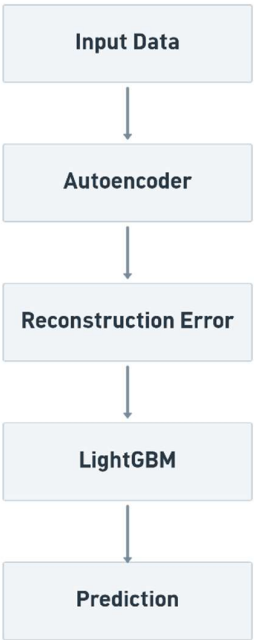


Figure 4.4: Hybrid Model Architecture

The hybrid model's precision was 0.53, its recall was 0.89, and its F1 score was 0.66. This is more precise than the standalone Autoencoder, and achieves high recall. But it is not as successful as the most successful supervised models.

4.10: Voting Ensemble

A voting approach was used to design a soft voting ensemble model using Random Forest and XGBoost classifiers. The final predictions were made by the ensemble using the mean (average) of the predicted probabilities produced by both models. This method was chosen because it takes the best features of bagging and boosting methods to make the classifier more robust.

Component	Description
Models	Random Forest, XGBoost
Voting Type	Soft Voting

Table 4.8: Voting Ensemble Components

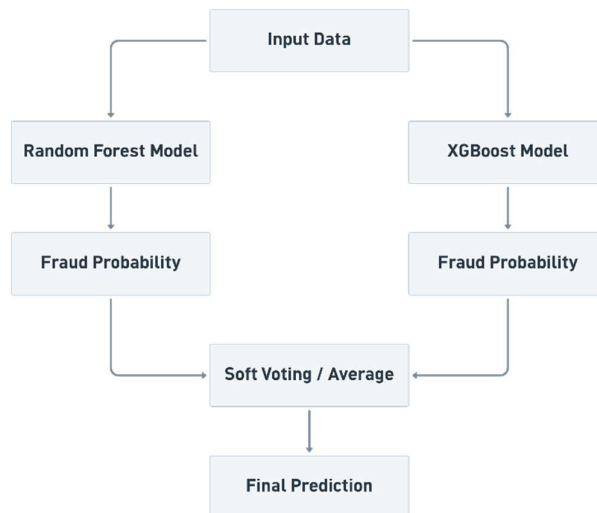


Figure 4.5: Voting Ensemble Workflow

The ensemble obtained the precision of 0.79, the recall of 0.86, and the F1 score of 0.82. This is a strong and stable performance, from both models, put together. The ensemble method gives enhanced robustness and gives a good trade-off between precision and recall.

4.11: Comparative Results

Model	Precision (Fraud)	Recall (Fraud)	F1-score (Fraud)
Logistic Regression	0.06	0.92	0.11
Random Forest	0.85	0.84	0.84
XGBoost	0.64	0.89	0.74
LightGBM	0.49	0.87	0.62
MLP	0.76	0.84	0.8
Autoencoder	0.27	0.83	0.41
Hybrid Model	0.53	0.89	0.66
Voting Ensemble	0.79	0.86	0.82

Table 4.9: Comparative Model Results

4.12: Model Performance Visualisation

4.12.1: Confusion Matrix Analysis

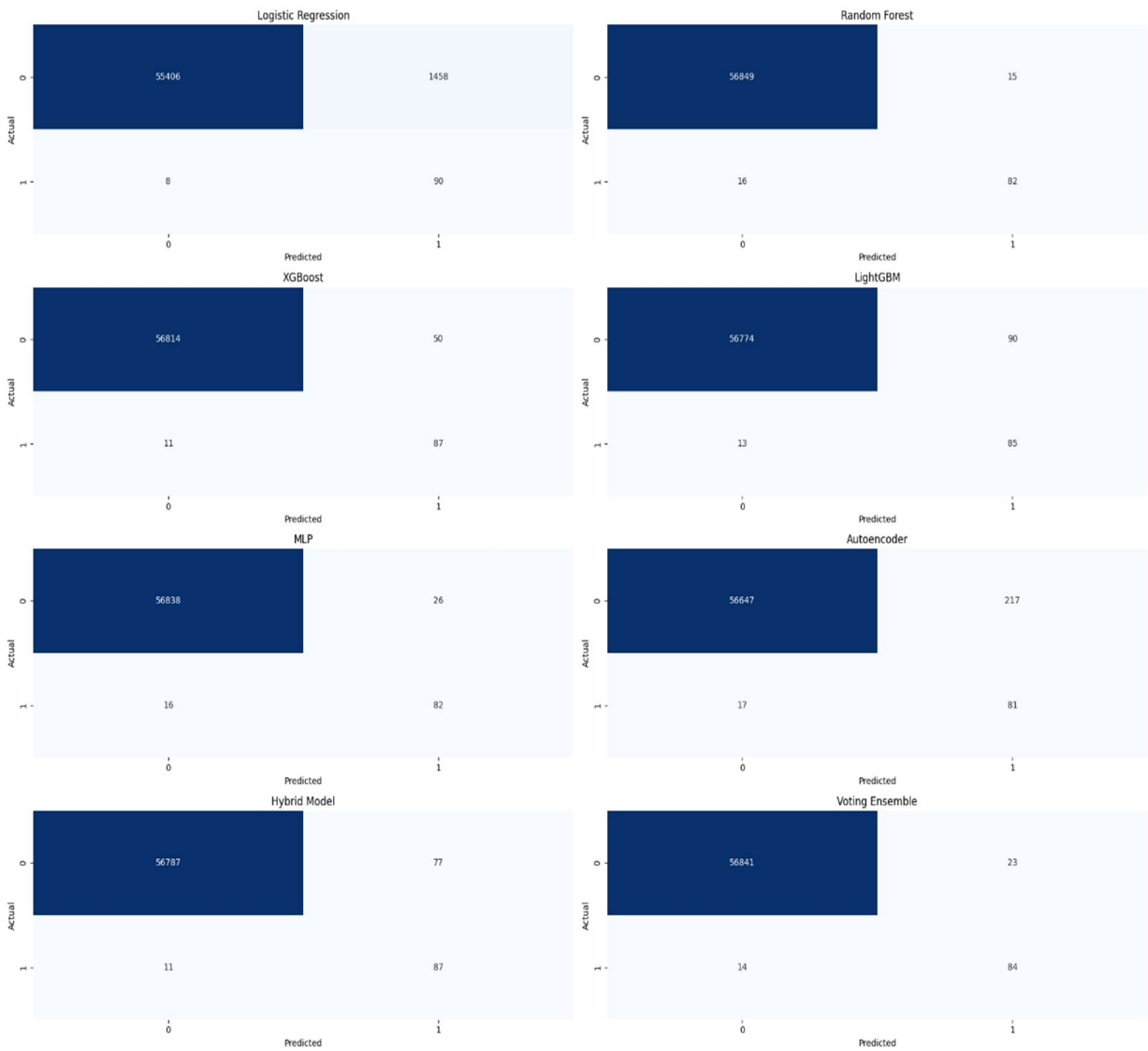


Figure 4.6: Confusion Matrices for Selected Models

The confusion matrixes give details of the classification results. The Random Forest and Voting Ensemble models have shown good performance, with a low rate of false positives and a high rate of correct identification of fraudulent transactions.

On the other hand, models like Logistic Regression produce higher numbers of false positives, which implies less performance in classification.

4.12.2: Model Comparison Using F1-Score

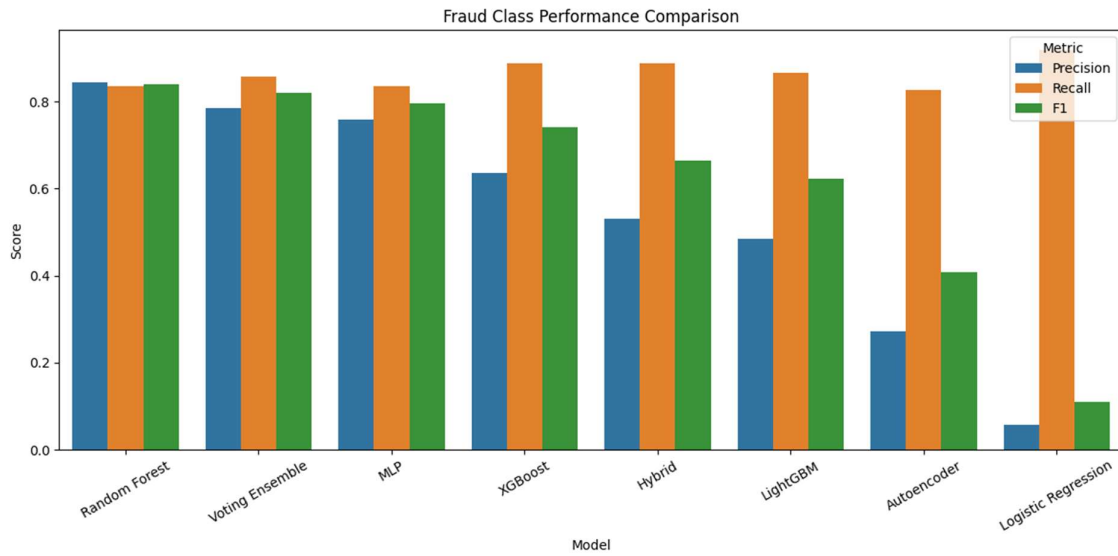


Figure 4.7: F1-Score Comparison of Models

The F1 score comparison reflects the overall performance of each model. The Voting Ensemble and Random Forest have the best overall F1-scores, meaning they are both precise and recalling.

So models like Autoencoder, which have relatively high recall, but lower precision, end up with low F1 scores.

4.12.3: ROC-AUC and Precision–Recall AUC Analysis

In addition, ROC-AUC and Precision–Recall AUC visualisations were produced to further assess the performance of the models. These metrics are more appropriate to measure the classification results on highly imbalanced datasets where accuracy can give misleading results.

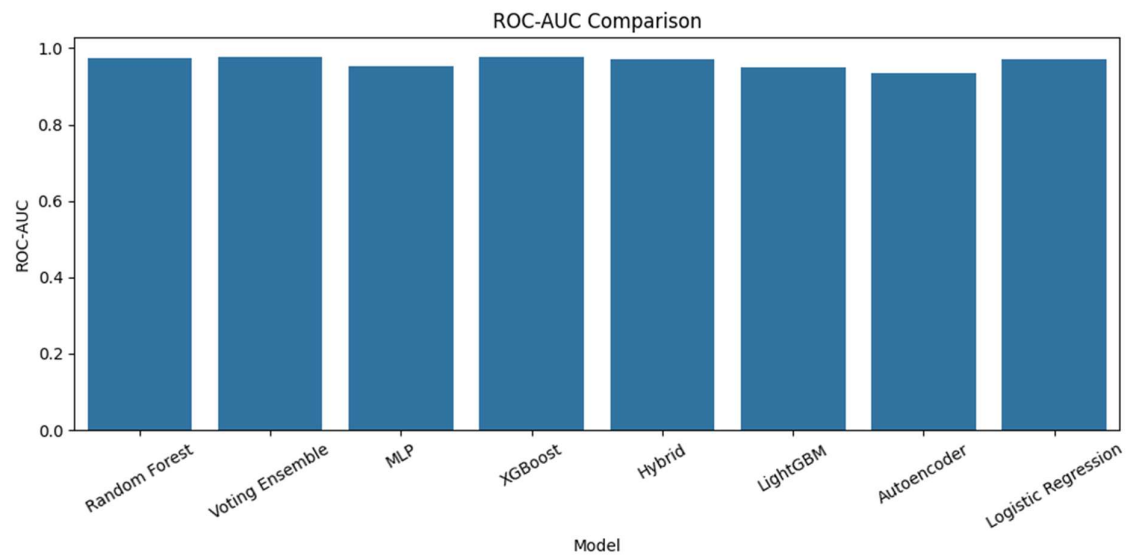


Figure 4.8: ROC-AUC Comparison

The ROC-AUC comparison shows how well each model performs at various classification thresholds and whether there are any similarities or differences in their performance. Random Forest, Voting Ensemble and MLP had the best ROC-AUC scores, making it have good discriminatory ability and stable classification performance.

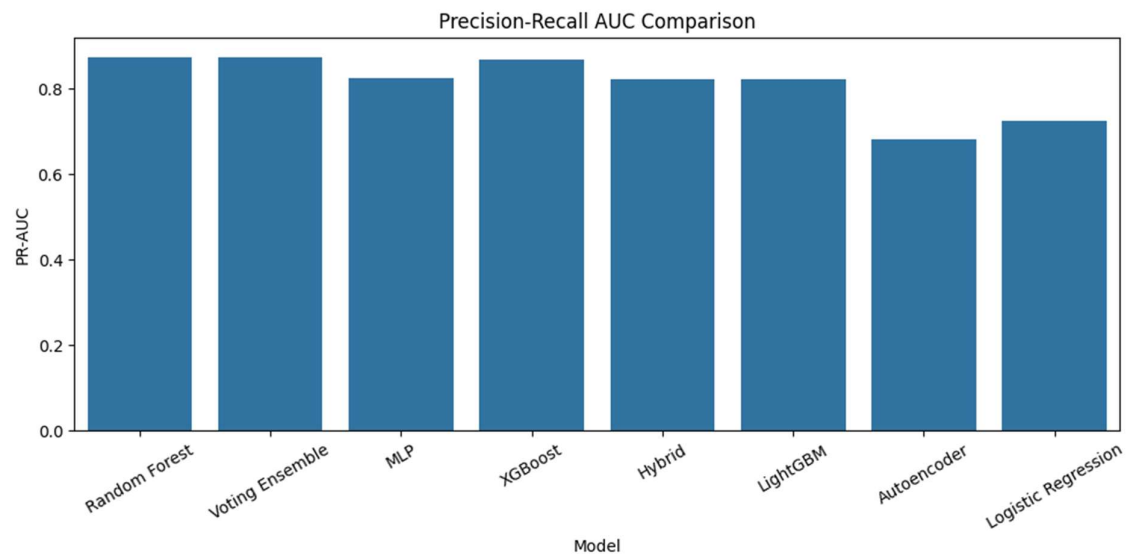


Figure 4.9: Precision–Recall AUC Comparison

The Precision–Recall AUC comparison gives the deeper insight in terms of the discrimination of the fraud class, which is the minority class in case of fraud detection. The models that performed better in terms of Precision–Recall were ensemble-based models, which showed a better trade-off between the correct identification of fraud cases and the reduction of false positives. On the other hand, models like Logistic Regression and Autoencoder have lower Precision–Recall performance, because they have a low precision despite high recall.

4.12: ROC Curves for Top Performing Models

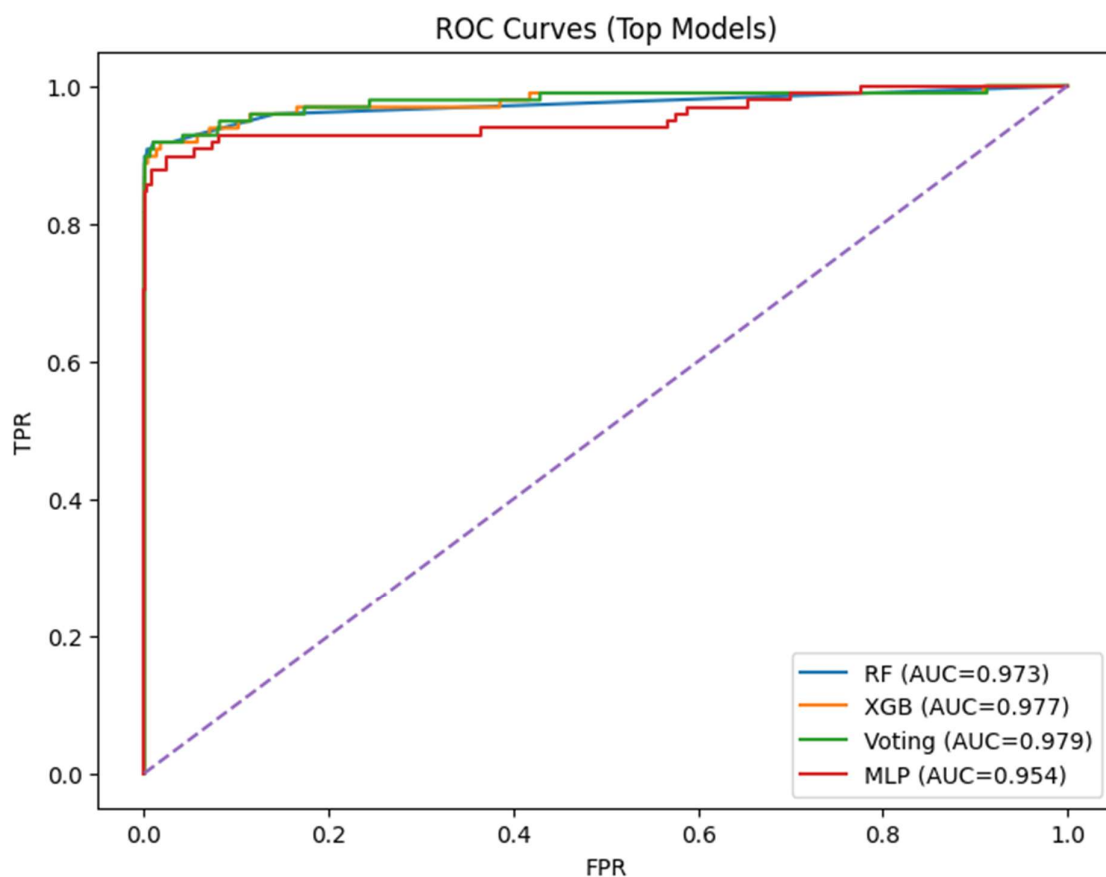


Figure 4.10: ROC Curves for Top Performing Models

The ROC curve plots of the best models also demonstrate the benefits of ensemble learning techniques. Random Forest and the Voting Ensemble were the classifiers with the curves that were most close to the upper left corner of the graph, showing the best classification performance for a number of threshold values. The results confirm the fact that ensemble

methods are best suited to detect credit card fraud in highly imbalanced datasets and yield the best and balanced performance.

4.13: Analysis of Model Behaviour

The results in the performance of the different models can be understood to a large extent as a result of the properties of the data set and learning mechanism of each algorithm. The imbalanced nature of the data, and the complexity and non-linearity of the fraud patterns, made models capable of identifying complex interactions between the features more effective than those that were simpler and linear. The results found in this study are also related to the existing studies on credit card fraud detection, where studies regarding the effectiveness of ensemble learning methods with imbalanced financial datasets are generally in line with the findings in this research.

Logistic Regression had a very high recall but very low precision rate for fraud class. This behaviour suggests that this model predicted that there were many fraud transactions that were not going to be caught, so it marked them as frauds, which reduced the number of false negatives. This made it easier to detect, but also led to a large proportion of false positives. The model's performance is also limited by the fact that its decision boundary is linear, which is not suitable for the financial transaction data, which has more complex non-linear relationships. Varmedja et al. (2019) also noted the same disadvantages of Logistic Regression, stating that traditional linear models lack the ability to provide a balanced performance in the detection of frauds in absence of advanced preprocessing and sampling techniques.

Random Forest had the highest F1 Score. The ensemble structure, in which many decision trees learn different interactions of features and vote in certain patterns, may also explain this good performance. Random Forest is also very robust to noise and overfitting and works well in highly non-linear and imbalanced datasets. This was because it is able to find the non-linear patterns which is why it far outperformed Logistic Regression. This is in line with the findings of Khalid et al., (2024) which showed that ensemble-based classifiers perform better than individual classifiers for fraud detection applications as it enhances the stability of the classification and generalisation.

XGBoost also performed very well, with a high score in Recall. The boosting mechanism in the model focuses on misclassified observations in an iterative fashion, thereby improving the ability to detect challenging cases of fraud. The accuracy of this model is slightly poorer than

the Random Forest model, however, which may mean that the model is prone to over-predicting fraud, causing extra false positives. In boosting algorithms this is the usual behaviour as they try to minimize classification errors while training. Taha and Malebary (2020) also noted that gradient boosting models had a high detection rate of frauds but did need careful tuning otherwise high false positive rates

In comparison, LightGBM performed better on recall and worse on precision than other supervised ensemble models. While LightGBM is efficient on large-scale data, it is limited by its leaf-wise tree growing approach which may result in over-splitting and excessive sensitivity to under-represented classes. This is probably the reason why more "false-positive" results were observed in the results. Taha and Malebary (2020) also mentioned that although LightGBM can be highly predictive, it can also be greatly affected by model tuning and class balancing techniques on imbalanced fraud data.

The Multi-Layer Perceptron (MLP) performed well and was consistent. Neural networks can learn complex nonlinear relationships in a "hidden layer representation" which makes them appropriate to fraud detection problems. The MLP failed to outperform the Random Forest though, possibly because the data consists mostly of structured tabular data, for which deep learning generally has not been very successful. Alharbi et al. (2022) found that deep learning models can yield competitive performance, but are not necessarily superior to ensemble methods in structured transaction data.

The Autoencoder had a high recall and low precision. The normal transactions were reconstructed very accurately by the Autoencoder whereas the fraudulent transactions had higher reconstruction errors and thus were detected as frauds. But not every odd transaction is malicious, leading to a higher false positive rate. This behaviour is typical of anomaly detection models, which are typically better for detecting suspicious activity than for accurate classifications. Jiang et al. (2023) also reported that anomaly detection techniques are capable of identifying abnormal transactions from unseen or unmarked fraud transactions, but may be less accurate because of legitimate anomalous transactions.

The Hybrid Model consisting of Autoencoder reconstruction error + LightGBM performed better than Autoencoder. The supervised model was able to use labelled training data as well as anomaly-related information, because the reconstruction error was added as an extra feature. The hybrid approach, however, did not outperform Random Forest or Voting

Ensemble methods, indicating that supervised ensemble learning is better in general for this dataset. The results are consistent with the results presented by Mienye and Sun (2023a) in which they showed that the integration of anomaly detection and supervised learning can enhance the performance of fraud detection while the ensemble classifiers generally showed improved overall classification balance.

The Voting Ensemble had one of the best performances on all evaluation measures. The ensemble leveraged the benefits of both the bagging and boosting techniques by using the predictions from both the Random Forest and XGBoost models. This minimized the effect of individual model weaknesses and improved the prediction stability. The ensemble's high balanced precision and recall values suggest that integrating multiple high performing classifiers in a fraud detection system can yield better system robustness. Mim et al. (2024) also found that by combining the strengths of various classifiers, soft voting ensemble methods significantly boost fraud detection results.

Overall results indicate that supervised ensemble models are preferable to detect credit card fraud in highly imbalanced tabular data. Anomaly detection techniques can be used for detecting suspicious behaviour, but they can be prone to higher FPR if applied as stand-alone solutions. Additionally, the study highlights the importance of model selection, preprocessing approaches, and methods for dealing with imbalanced data in financial data to ensure accurate fraud detection results in practical applications.

Chapter 5: Conclusion and Future Work

5.1: Introduction

This chapter provides a brief overview of the main results, their performance, difficulties and limitations, and areas for further research and development. The chapter examines the effectiveness of the financial fraud detection framework that has been developed, and its applicability to real financial systems.

5.2 Summary of the Study

The primary objective of this study was to create and test machine learning techniques for fraudulent credit card transactions in a highly imbalanced dataset. This is achieved by implementing several supervised learning, deep learning and anomaly detection models and comparing them with relevant evaluation metrics. The study adopted the CRISP-DM model as a framework for understanding the data, pre-processing, modeling, evaluation and analysis. The Exploratory Data Analysis identified some key features of the dataset such as significant class imbalance, non-uniform distributions in the transactions and non-linear relationships between the features. The results affected the preprocessing approach and aided in the implementation of better machine learning techniques. The imbalance problem was addressed by using the SMOTE technique on the training data. A variety of models were evaluated not just accuracy, but precision, recall, F1 score, and ROC-AUC. The models used were Logistic Regression, Random Forest, XGBoost, LightGBM, Multi-Layer Perceptron (MLP), Autoencoder, Hybrid Autoencoder-LightGBM model, and the Voting Ensemble model.

5.3 Summary of Key Findings

The results demonstrated that there was a clear difference in the performance of the models. As expected, traditional linear models such as Logistic Regression achieved very high recall but low precision, meaning that they had difficulty separating fraudulent transactions from legitimate ones. This was a validation of the nonsimplicity of fraud patterns in the fraud detection dataset. Random Forest showed the overall best performance among the supervised learning methods with good values of precision, recall and F1 score. The model was able to identify non-linear relationships in the data and was found to be robust against overfitting and noise. XGBoost and LightGBM were also good performers and particularly excelled in fraud detection. XGBoost exhibited high recall with good classification ability, and LightGBM had

good efficiency in training. Both models were not as precise as the Random Forest though, resulting in more false positives. The Multi-Layer Perceptron demonstrated reasonably good performance with stable and competitive results, demonstrating the feasibility of learning complex transaction patterns using neural networks. Even so, the deep learning approach did not perform better than the best of the ensemble models, which is probably because the data was not very high-dimensional but mostly structured tabular data. The Autoencoder obtained a high Recall and a low Precision, as is typical of anomaly detection models. It successfully recognized abnormal transactions but a lot of abnormal transactions were not frauds which led to more false positives. The Hybrid Model was able to outperform the standalone Autoencoder, whereas it did not outperform the best ensemble models. The Voting Ensemble model performed very well and was pretty balanced, with high accuracy and reliability due to the advantage of the two models, Random Forest and XGBoost. This method improved classification stability and decreased the weakness of each model and is one of the most effective methods evaluated in this study. Overall, the findings suggest that supervised ensemble learning methods are the best choice for credit card fraud detection on highly imbalanced tabular datasets

5.4 Model Performance Evaluation

Random Forest and the Voting Ensemble were the best for the balance between fraud detection capability and control of false positives according to the comparative evaluation. The models consistently performed well in terms of F1-scores and ROC-AUC values, showing good classification performance at various thresholds. Recall was also given more importance by models such as Logistic Regression, and Autoencoder, which produce too many false positives even if they detect many fraudulent transactions. In the realm of fraud detection, high recall is critical but an excess of false alarms may restrict the usefulness of these systems in practical financial environments. This was confirmed by visualization results. ROC-AUC and Precision-Recall AUC analyses showed that ensemble models performed well in classification for different thresholds. The confusion matrix analysis also revealed that ensemble methods performed better than baseline and anomaly detection methods in terms of the number of classification errors. These findings are consistent with the earlier work that shows that ensemble learning methods can outperform single machine learning methods and the unsupervised anomaly detection method in fraud detection.

5.5 Challenges and Limitations

There were a number of difficulties in conducting this research. The most difficult task was the extreme class imbalance in the data, in which the number of fraudulent transactions was a very small proportion of the total observations. This imbalance complicated training models to accurately identify fraud without causing too many false positives. Another restriction was that the data were anonymous. Most of the features were transformed using PCA, which led to the loss of original meanings from the variables. This made it difficult to gain deeper understanding of features and behaviour within the domain. Furthermore, the data was collected from one financial environment at a specific period of time and results may not be generalizable to all financial environments or evolving patterns of financial fraud. Fraud activity is a constantly changing landscape, and fraud models based on old fraud patterns will not perform well when presented with new ones. Finally, this project did not cover full real-time deployment and integration in a live financial system, and therefore the performance of the models in real-world transactions was not evaluated

5.6 Future Work and Improvements

There are several opportunities to extend and enhance this research. For a more detailed analysis of the transaction patterns, advanced deep learning models such as the Transformer network, Graph Neural Networks (GNNs), and Long Short-Term Memory (LSTM) networks could be explored in future research. The other key research direction in which the technology needs to be expanded is real-time streaming analytics. The use of machine learning models would be more useful in operational financial systems if fraud detection systems could process real time transactions. Future research could also incorporate the performance of the models on other datasets from other financial institutions and other regions to improve generalizability and robustness.

5.7 Conclusion

This study shows that machine learning can be successfully used to identify fraudulent credit card transactions in highly imbalanced data. Results showed that the supervised learning models based on ensembles such as Random Forest and Voting Ensemble performed the best regarding the evaluation metrics with the best balance between them. Anomaly detection techniques like Autoencoders performed well in detecting the suspicious pattern, but there was a higher rate of false positive. Hence, it is concluded that supervised ensemble learning is the most suitable method for fraud detection in structured transaction data set. The research

was successful in accomplishing its objectives by the development and testing of various fraud detection methods in a cohesive analytical framework. The models developed and the results obtained will add to the ever-expanding body of financial fraud analytics and demonstrate the utility of machine learning in enhancing fraud detection systems in today's digital financial landscape.

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